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(45) Regular Expression

Regular Expressions are used for representing certain sets of strings in an algebraic function. fashion.

$a, b, c \rightarrow \Sigma$

- 1) Any terminal symbol i.e. symbols $\in \Sigma$ including Λ and ϕ are regular expressions.
- 2) The Union of two regular expressions is also a regular expression $R_1, R_2 (R_1 + R_2)$
- 3) The Concatenation of two regular expressions is also a regular expression. $(R_1 R_2)$
- 4) The iteration (or Closure) of a regular expression is also a regular expression $R \rightarrow R^+$
 $a^+ = \{a, aa, aaa, \dots\}$
- 5) The regular expression over Σ are precisely those obtained ~~req~~ recursively by the application of the above rules once or several times,



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(46) Regular Expression - Examples

1) $\{0, 1, 2\}$ 0 or 1 or 2.

$$R = \underline{0 + 1 + 2}$$

2) $\{\lambda^E, a b\}$

$$\lambda = \epsilon$$

$$R = \lambda^E + ab$$

3) $\{ab b, a, b, b b a\}$

$$R = ab b + a + b + b b a$$

4) $\{\lambda^E, 0, 00, 000, \dots\}$ closure of 0

$$\Rightarrow R = 0^*$$

5) $\{1, 11, 111, 1111, \dots\}$ closure of 1

$$R = 1^+ \quad +: \text{not } \epsilon$$

(47) Identities of Regular Expression

$$1) \phi + R = R$$

$$7) RR^* = R^*R$$

$$2) \underline{\phi R + R \phi = \phi}$$

$$8) (R^*)^* = R^*$$

$$\epsilon + R^+ = R^+$$

$$3) \epsilon R = R \epsilon = R$$

$$9) \epsilon + RR^* = \epsilon + R^*R = R^*$$

$$4) \underline{\epsilon^* = \epsilon \text{ and } \phi^* = \epsilon}$$

$$10) (RQ)^*P = P(QP)^*$$

$$5) R + R = R$$

$$11) (P + Q)^* = (P^*Q^*)^*$$

$$6) R^*R^* = R^*$$

$$= (P^* + Q^*)^*$$

$$12) (P + Q)R = PR + QR \text{ and}$$

$$R(P + Q) = RP + RQ$$